

1. Review differentiation (product rule, quotient rule and chain rule)
2. Equation of a tangent line at a given point $x=a$ $y - f(a) = f'(a)(x - a)$
3. Integration u-sub, by parts, trig substitution and partial fractions decomposition with long division of polynomials
4. Evaluating improper integrals (three types of improper integrals) using limits
5. Comparison test for improper integrals
6. Numerical integration Mid point rule, left-sum, right-sum, trapezoidal rule and Simpsons rule.

Derivatives:

Example:

Differentiate

1) $\frac{d}{dx} [\sqrt{x} \cdot \sec(2x)]$

$f = \sqrt{x}$ $g = \sec(2x)$
 $f' = \frac{1}{2\sqrt{x}}$ $g' = 2 \cdot \sec(2x) \tan(2x)$

$$\left[\frac{\sec(2x)}{2\sqrt{x}} + \sqrt{x} \cdot 2 \sec(2x) \tan(2x) \right]$$

Short cuts:

$$\frac{d}{dx} [\cos(ax)] = -a \sin(ax)$$

$$\frac{d}{dx} [e^{ax}] = a e^{ax}$$

$$2) \frac{d}{d\theta} \left[\frac{\ln(\theta)}{\tan(3\theta)} \right] = \frac{\tan(3\theta) \cdot \frac{1}{\theta} - \ln(\theta) \cdot 3\sec^2(3\theta)}{(\tan(3\theta))^2}$$

Quotient rule

$f = \ln(\theta)$	$g = \tan(3\theta)$
$f' = \frac{1}{\theta}$	$g' = 3\sec^2(3\theta)$
$\frac{g \cdot f' - f \cdot g'}{g^2}$	

$$3) \frac{d}{ds} \left[\underbrace{(s^2 - 2)}_f \underbrace{e^{-s+5}}_g \right] = 2s \cdot e^{-s+5} - (s^2 - 2)e^{-s+5}$$

$f = s^2 - 2$	$g = e^{-s+5}$
$f' = 2s$	$g' = -e^{-s+5}$

(Note: The original image shows a crossed-out assignment of f and g, indicating a correction or a specific choice for the product rule.)

$$4) \frac{d}{dt} \left[\frac{\cot(5t)}{t+1} \right] = \frac{(t+1) \cdot (-5\csc^2(5t)) - \cot(5t) \cdot 1}{(t+1)^2}$$

$$5) \frac{d}{dx} \left[\int_{\sqrt{x}}^{\cos(7x)} \ln(t-1) dt \right] \leftarrow \text{The Fundamental Theorem of Calculus (part I) applies}$$

$$= \ln(\cos(7x) - 1) \cdot \frac{d}{dx} [\cos(7x)] - \ln(\sqrt{x} - 1) \cdot \frac{1}{2\sqrt{x}}$$

$$= \ln(\cos(7x) - 1) \cdot (-7\sin(7x)) - \ln(\sqrt{x} - 1) \cdot \frac{1}{2\sqrt{x}}$$

$$6) \frac{d}{dx} \left[\int_{5^x}^{100} \tan(\theta) d\theta \right] \rightarrow \text{Recall } \int_a^b f(x) dx = - \int_b^a f(x) dx$$

$$\frac{d}{dx} \left[- \int_{100}^{5^x} \tan(\theta) d\theta \right] = - \left[\tan(5^x) \cdot \frac{d}{dx} [5^x] - \tan(100) \cdot \frac{d}{dx} [100] \right]$$

Recall: $\frac{d}{dx} [a^x] = a^x \cdot \ln(a)$

$$= - \tan(5^x) \cdot 5^x \cdot \ln(5)$$

$$\begin{aligned}
 7) \quad \frac{d}{dx} \left[\int_{\sec(2x)}^3 \sqrt{t} \cdot e^t dt \right] &= \frac{d}{dx} \left[- \int_3^{\sec(2x)} \sqrt{t} \cdot e^t dt \right] \\
 &= - \sqrt{\sec(2x)} \cdot e^{\sec(2x)} \cdot \frac{d}{dx} [\sec(2x)] \\
 &= - \sqrt{\sec(2x)} \cdot e^{\sec(2x)} \cdot 2 \cdot \sec(2x) \cdot \tan(2x)
 \end{aligned}$$

its derivative is constant

8) Find equation of a tangent line to $f(x) = \sin(3x)$ at $x = \frac{\pi}{4}$.

Equation of a tangent line formula @ $x = \pi/4$

$$y - f\left(\frac{\pi}{4}\right) = f'\left(\frac{\pi}{4}\right) \left(x - \frac{\pi}{4}\right)$$

$$f\left(\frac{\pi}{4}\right) = \sin\left(3\left(\frac{\pi}{4}\right)\right) = \frac{\sqrt{2}}{2}$$

$$f'(x) = \frac{d}{dx} [\sin(3x)] = 3 \cdot \cos(3x) \Big|_{x=\frac{\pi}{4}} = 3 \cos\left(3 \cdot \frac{\pi}{4}\right) = 3 \cdot \left(-\frac{\sqrt{2}}{2}\right) = -\frac{3\sqrt{2}}{2}$$

$$\boxed{y - \frac{\sqrt{2}}{2} = -\frac{3\sqrt{2}}{2} \left(x - \frac{\pi}{4}\right)} \checkmark$$

Examples:

$$1) \int \frac{e^x}{1+e^{2x}} dx = \int \frac{1}{1+(e^x)^2} \cdot \overbrace{e^x}^{du} dx$$

u-sub
 $u = e^x$
 $du = e^x dx$

$$= \int \frac{1}{1+u^2} \cdot du = \tan^{-1}(u) + C = \boxed{\tan^{-1}(e^x) + C}$$

$$2) \int \frac{dt}{t\sqrt{\ln(t)}} = \int \frac{1}{\sqrt{\ln t}} \cdot \overbrace{\frac{1}{t}}^{du} dt$$

$u = \ln t$
 $du = \frac{1}{t} dt$

$$\int \frac{1}{\sqrt{u}} \cdot du = 2\sqrt{u} + C = \boxed{2\sqrt{\ln(t)} + C}$$

Short cuts:

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C$$

$$\int \sin(ax) dx = -\frac{1}{a} \cos(ax) + C$$

$$\int \sec^2(ax) dx = \frac{1}{a} \tan(ax) + C$$

3)

When we apply u-sub for definite integrals, we must change the bound of integration. Otherwise, integrate aside without the bound, as if it were an indefinite integral.

$$\int_0^1 \frac{e^x + 1}{e^x + x} dx = \int_1^{e+1} \frac{du}{u} = \ln|u| \Big|_1^{e+1} = \ln|e+1| - \ln|1| = \boxed{\ln|e+1|}$$

0

u-sub.
 $u = e^x + x$
 $du = e^x + 1 dx$

Bounds:

if $x=0$, $u = e^0 + 0 = 1$
 if $x=1$, $u = e^1 + 1$

$$4) \int_2^{e^4} \frac{dx}{x \sqrt{\ln x}} = \int_{\ln(2)}^4 \frac{1}{\sqrt{u}} du = 2\sqrt{u} \Big|_{\ln(2)}^4 = 2\sqrt{4} - 2\sqrt{\ln(2)} = \boxed{4 - 2\sqrt{\ln(2)}}$$

$u = \ln x \rightarrow$ Bounds:
 if $x=2$, $u = \ln(2)$
 if $x=e^4$, $u = \ln(e^4) = 4 \ln(e) = 4$

1
1

5) Tricky u-sub.

$$\int_1^2 2x \sqrt{x-1} dx = \int_0^1 2 \cdot (u+1) \sqrt{u} du =$$

u-sub

$$u = x - 1$$

$$du = dx$$

Solve for "x"

$$x = u + 1$$

→ Bounds:

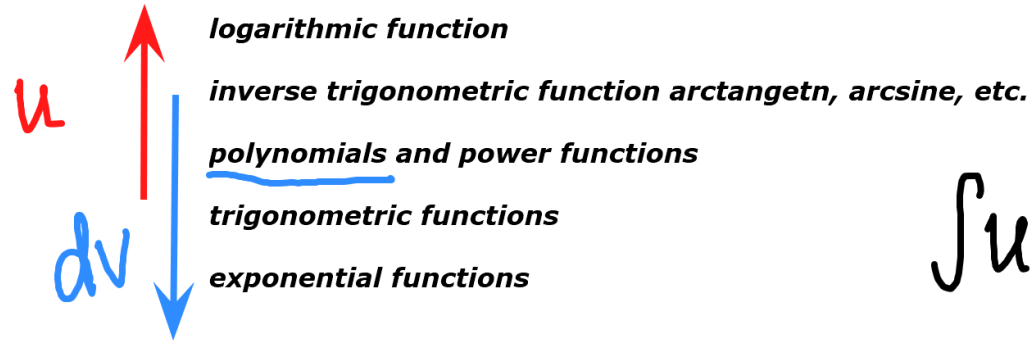
$$\text{if } x = 1, u = 0$$

$$\text{if } x = 2, u = 1$$

$$\begin{aligned} \rightarrow &= \int_0^1 2u^{3/2} + 2u^{1/2} du = \frac{4}{5}u^{5/2} + \frac{4}{3}u^{3/2} \Big|_0^1 = \frac{4}{5}(1)^{5/2} + \frac{4}{3}(1)^{3/2} - 0 \\ &= \boxed{\frac{32}{15}} \end{aligned}$$

Integration by parts

The list of function to consider while labeling the given integral in terms of u and dv



$$\int u dv = uv - \int v du$$

Examples:

Integrate:

1) $\int \underbrace{3x}_u \underbrace{e^{5x}}_{dv} dx$

By parts:

$$u = 3x$$
$$du = 3 dx$$

$$dv = e^{5x} dx$$
$$v = \frac{1}{5} e^{5x}$$

Recall:

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C$$

$$\int u dv = uv - \int v du$$

$$3x \cdot \left(\frac{1}{5} e^{5x} \right) - \int \frac{1}{5} e^{5x} \cdot 3 dx$$

$$= \frac{3}{5} x e^{5x} - \frac{3}{5} \int e^{5x} dx = \frac{3}{5} x e^{5x} - \frac{3}{25} e^{5x} + C$$

$$2) \int \frac{\ln(x)}{\sqrt{x}} dx = \int \underbrace{\ln(x)} \cdot \underbrace{\frac{1}{\sqrt{x}} dx}$$

By parts:

$$\begin{aligned} u &= \ln(x) \\ du &= \frac{1}{x} dx \end{aligned} \quad \begin{aligned} dv &= \frac{1}{\sqrt{x}} dx \\ v &= 2\sqrt{x} \end{aligned}$$

$$\int u dv = uv - \int v du$$

$$\begin{aligned} 2\sqrt{x} \cdot \ln(x) - \int 2\sqrt{x} \cdot \frac{1}{x} dx &= 2\sqrt{x} \cdot \ln(x) - 2 \int \frac{1}{\sqrt{x}} dx \\ &= 2\sqrt{x} \cdot \ln(x) - 4\sqrt{x} + C \end{aligned}$$

3) By parts applied twice

$$\int \underbrace{t^2} \cdot \underbrace{e^{3t}} dt$$

$$\begin{aligned} u &= t^2 \\ du &= 2t dt \end{aligned} \quad \begin{aligned} dv &= e^{3t} dt \\ v &= \frac{1}{3} e^{3t} \end{aligned}$$

$$\int u dv = uv - \int v du$$

$$= \underbrace{\frac{t^2}{3} e^{3t}} - \underbrace{\frac{2}{3} \int t e^{3t} dt} \Rightarrow$$

We apply by parts again

$$\begin{aligned} u &= t \\ du &= dt \end{aligned} \quad \begin{aligned} dv &= e^{3t} dt \\ v &= \frac{1}{3} e^{3t} \end{aligned}$$

$$= \frac{t^2}{3} e^{3t} - \frac{2}{3} \left[\frac{t}{3} e^{3t} - \frac{1}{3} \int e^{3t} dt \right]$$

$$= \frac{t^2}{3} e^{3t} - \frac{2}{3} \left[\frac{t}{3} e^{3t} - \frac{1}{9} e^{3t} \right] + C$$

$$= \frac{t^2}{3} e^{3t} - \frac{2t}{9} e^{3t} + \frac{2}{27} e^{3t} + C$$

$$\boxed{1 = 1 \cdot x^0}$$

4) Integration by parts along with u-sub.

$$\int \sin^{-1} x \, dx = \int \underbrace{\sin^{-1} x}_u \cdot \underbrace{1 \, dx}_{dv}$$

$$\int u \, dv = uv - \int v \, du$$

$$x \cdot \sin^{-1}(x) - \int \frac{x}{\sqrt{1-x^2}} \, dx =$$

$$\begin{array}{ll} u = \sin^{-1}(x) & dv = 1 \, dx \\ du = \frac{1}{\sqrt{1-x^2}} \, dx & v = x \end{array}$$

$$\begin{array}{l} \text{u-sub.} \\ u = 1-x^2 \\ du = -2x \, dx \\ -\frac{du}{2} = x \, dx \end{array}$$

$$= -\frac{1}{2} \int \frac{1}{\sqrt{u}} \, du = -\frac{1}{2} \cdot 2\sqrt{u} + C = -\sqrt{u} + C$$

$$\Rightarrow \boxed{x \cdot \sin^{-1}(x) + \sqrt{1-x^2} + C}$$

! $\int \tan^{-1}(x) \, dx$

5)

Given a twice-differentiable function f , evaluate

$$\int_0^2 \underbrace{x}_{u} \cdot \underbrace{f''(x)}_{dv} dx \quad \text{if}$$

By parts

$$\begin{aligned} u &= x & dv &= f''(x) dx \\ du &= dx & v &= f'(x) \end{aligned}$$

x	$f(x)$	$f'(x)$	$f''(x)$
0	3	-1	2
2	3	-1	4

$$\int u dv = uv - \int v du$$

$$x \cdot f'(x) - \int f'(x) dx$$

$$= x \cdot f'(x) - f(x) + C$$

Thus,

$$\int_0^2 x \cdot f''(x) dx = x \cdot f'(x) - f(x) \Big|_0^2 = \underbrace{2 \cdot f'(2)}_{-1} - \underbrace{f(2)}_3 - \left(\underbrace{0 \cdot f'(0)}_0 - \underbrace{f(0)}_3 \right) = \boxed{-2}$$

Integrating Rational Functions

Recall: A rational function is a ratio of two polynomials, so when we need to integrate a rational function **FIRST THING WE COMPARE THE DEGREES!** We compare the highest exponents in the numerator and the denominator.

If the degree in the numerator EQUAL or GREATER than the degree in the denominator, we perform the long division for polynomials

$$\frac{P(x)}{D(x)} = Q(x) + \frac{R(x)}{D(x)}$$

If the degree in the numerator is SMALLER than the degree in the denominator, we factor the denominator polynomial and perform partial fraction decomposition, so that we get the sum or difference of simpler fractions (rations) which are easier to integrate.

Examples:

1) Rewrite the function for the partial decomposition.

$$\frac{2x^6 + 18x^2 + 30x + 9}{x^3 \cdot (x-1)^2 \cdot (x^2+4)^2 \cdot (x-8)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x^3} + \frac{D}{x-1} + \frac{E}{(x-1)^2} + \frac{Fx+G}{x^2+4} + \frac{Hx+I}{(x^2+4)^2} + \frac{J}{x-8}$$

degree 10

Remark $x^3 = (x-0)(x-0)(x-0)$
this is repeated linear factor, so it is case II

2) Evaluate $\int_{-1}^0 \frac{x^3 - 4x + 1}{x^2 - 3x + 2} dx$ ← Long Division applies

$$\begin{array}{r}
 x^2 - 3x + 2 \overline{) x^3 + 0x^2 - 4x + 1} \\
 \underline{-(x^3 - 3x^2 + 2x)} \\
 3x^2 - 6x + 1 \\
 \underline{-(3x^2 - 9x + 6)} \\
 3x - 5
 \end{array}$$

Diagram illustrating the long division process with arrows showing the subtraction of the divisor from the dividend and the resulting remainder.

$$\frac{P(x)}{D(x)} = \underbrace{Q(x)}_{\text{Quotient polynomial}} + \frac{\overbrace{R(x)}^{\text{remainder}}}{\underbrace{D(x)}_{\text{divisor}}}$$

← this is the remainder

$$\int \frac{x^3 - 4x + 1}{x^2 - 3x + 2} dx = \int x + 3 + \frac{3x-5}{x^2-3x+2} dx = \int x + 3 + \frac{1}{x-2} + \frac{2}{x-1} dx \Rightarrow$$

Partial fractions applies

$$\frac{3x-5}{(x-2)(x-1)} = \frac{A \cdot x^{-1}}{x-2} + \frac{B \cdot x^{-2}}{x-1} = \frac{1}{x-2} + \frac{2}{x-1}$$

two linear distinct factors (case I)

$$3x-5 = A(x-1) + B(x-2)$$

$$\text{if } x=1 \rightarrow 3(1)-5 = A \cdot 0 + B(-1) \\ -2 = -B, \text{ so } \boxed{B=2}$$

$$\text{if } x=2 \rightarrow 3(2)-5 = A(1) + B \cdot 0 \\ \boxed{1=A}$$

$$= \frac{x^2}{2} + 3x + \ln|x-2| + 2\ln|x-1| + C$$

Once it is integrated, we evaluate it because the original integral is the definite integral.

Thus,

$$\begin{aligned} \int_{-1}^0 \frac{x^3 - 4x + 1}{x^2 - 3x + 2} dx &= \frac{x^2}{2} + 3x + \ln|x-2| + 2\ln|x-1| \Big|_{-1}^0 \\ &= 0 + 0 + \ln|0-2| + 2\ln|0-1| - \left(\frac{(-1)^2}{2} + 3(-1) + \ln|-1-2| + 2\ln|-1-1| \right) \\ &= \boxed{\ln|-2| - \frac{1}{2} + 3 - \ln|-3| - 2\ln|-2|} \end{aligned}$$

$$3) \int \frac{2x^2 - x - 1}{(x^2 + 1)(x - 2)} dx$$

Degree in the denominator is 3, which is greater than the degree in the numerator. Partial fractions decomposition applies.

$$\frac{2x^2 - x - 1}{(x^2 + 1)(x - 2)} = \frac{Ax + B}{x^2 + 1} + \frac{C}{x - 2}$$

(Red annotations: $x-2$ above the second term, x^2+1 above the first denominator)

an irreducible quadratic, so Case III

$$2x^2 - x - 1 = (Ax + B)(x - 2) + C(x^2 + 1)$$

We expand this side, so we will be matching the coefficients on the left-hand side

$$2x^2 - x - 1 = Ax^2 + Bx - 2Ax + Cx^2 + C - 2B$$

$$2x^2 - x - 1 = (A + C)x^2 + (B - 2A)x + C - 2B$$

$2 = A + C$
 $-1 = B - 2A$
 $-1 = C - 2B$



Solve for A, B and C.

$$A + C = 2$$

$$B - 2A = -1$$

$$C - 2B = -1$$

$$A = \frac{B+1}{2}$$

$$\Rightarrow \boxed{A=1}$$

$$\hookrightarrow C = 2B - 1$$

$$\boxed{C=1}$$

$$\frac{B+1}{2} + 2B - 1 = 2$$

$$B + 1 + 4B - 2 = 4$$

$$5B = 5 \rightarrow \boxed{B=1}$$

$$\frac{Ax+B}{x^2+1} + \frac{C}{x-2} = \frac{x+1}{x^2+1} + \frac{1}{x-2} \Rightarrow$$

$$\int \frac{x+1}{x^2+1} + \frac{1}{x-2} dx = \int \frac{x}{x^2+1} + \frac{1}{x^2+1} + \frac{1}{x-2} dx =$$

$\underbrace{\hspace{1cm}}_{u\text{-sub}}$
 $u = x^2 + 1$
 $\frac{du}{2} = x dx \rightarrow \frac{1}{2} \int \frac{1}{u} du$

$$= \frac{1}{2} \ln|x^2+1| + \tan^{-1}(x) + \ln|x-2| + C$$

Completing the square method allows us to rewrite a quadratic function the standard form (vertex form), so that we can apply integral of arctangent.

Recall : $\int \frac{1}{1+x^2} dx = \tan^{-1} x + C$

$$\int \frac{1}{1+u^2} du = \tan^{-1}(u) + C$$

$$\int \frac{1}{a^2 + u^2} du = \frac{1}{a} \tan^{-1}\left(\frac{u}{a}\right) + C$$

If we see only an irreducible quadratic in the denominator (and no other terms), then we bring that quadratic function into standard form and apply the last formula.

$$\int \frac{1}{x^2 + 2x + 7} dx = \int \frac{1}{(x+1)^2 + 6} dx = \int \frac{1}{(\sqrt{6})^2 + \underbrace{(x+1)^2}_{u=x+1}} dx = \int \frac{1}{(\sqrt{6})^2 + u^2} du$$

$$x^2 + 2x + \left(\frac{2}{2}\right)^2 - \left(\frac{2}{2}\right)^2 + 7$$

$$\left(x + \frac{2}{2}\right)^2 - \left(\frac{2}{2}\right)^2 + 7 = (x+1)^2 + 6$$

$$u = x+1 \\ du = dx$$

$$= \frac{1}{\sqrt{6}} \tan^{-1}\left(\frac{u}{\sqrt{6}}\right) + C$$

$$= \frac{1}{\sqrt{6}} \tan^{-1}\left(\frac{x+1}{\sqrt{6}}\right) + C$$

$$\begin{aligned}
 \underline{3x^2 + 6x + 9} &= 3[x^2 + 2x + 3] \\
 &= 3\left[x^2 + 2x + \left(\frac{2}{2}\right)^2 - \left(\frac{2}{2}\right)^2 + 3\right] \\
 &= 3\left[\left(x + \frac{2}{2}\right)^2 - \left(\frac{2}{2}\right)^2 + 3\right] = 3\left[(x+1)^2 + 2\right] = 3(x+1)^2 + 6
 \end{aligned}$$

$$ax^2 + bx + c = a \cdot (x - h)^2 + k$$

$$\underline{A^2 + 2AB + B^2 = (A + B)^2}$$

Consider $\int \sqrt{3x^2 + 6x + 9} \, dx = \int \sqrt{3(x+1)^2 + 6} \, dx$

Which trigonometric substitution
can be applied here?

$$= \int \sqrt{3[(x+1)^2 + 2]} \, dx = \sqrt{3} \int \sqrt{(x+1)^2 + (\sqrt{2})^2} \, dx$$

tangent case.

$$\begin{aligned}
 & \left. \begin{array}{l} u = x+1 \\ du = dx \end{array} \right\} \sqrt{3} \int \sqrt{u^2 + (\sqrt{2})^2} \, du
 \end{aligned}$$

$$\begin{aligned}
 & x+1 = \sqrt{2} \tan \theta \quad \rightarrow \quad u = \sqrt{2} \tan \theta \\
 & \quad \quad \quad \rightarrow \quad x = \sqrt{2} \tan \theta - 1
 \end{aligned}$$